

Exploratory Data Analysis: Unveiling Trends in Food Delivery

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Title: Unveiling Trends in Food Delivery

Analyzing factors affecting food delivery performance.

Abstract:

We created a predictive model to forecast food delivery times more accurately. We cleaned and analyzed a Kaggle dataset to identify factors that impact delivery performance, such as vehicle type, weather, and traffic. The resulting model provides valuable insights to improve operational efficiency and customer satisfaction in the food delivery industry.

Introduction:

In the fast-paced world of online food delivery, the efficiency of service delivery directly impacts customer satisfaction and business success. Recognizing the need for enhanced predictability and speed in food delivery, A comprehensive analysis titled "Food Delivery Time Analysis." This project is dedicated to dissecting historical data from an extensive dataset provided by an online retail platform sourced from Kaggle. The primary objective is to refine the accuracy of delivery time forecasts, which is crucial for improving both the delivery process and the overall customer experience. This project involves collecting and loading data into an analytical framework with Pandas, followed by data preparation and cleaning to ensure accuracy.

We analyze the data to uncover variables that impact delivery times. This includes examining order type, delivery personnel details, and external factors like traffic and weather. We use visual techniques to highlight patterns and trends that affect delivery performance. We created a predictive model to improve delivery services. It considers various factors including vehicle types, weather conditions, city dynamics, and festive periods. The goal is to predict delivery times accurately and adapt to different delivery environments.

Methodology

Our methodology for the "Food Delivery Time Analysis" project included several key steps:

1. **Data Collection:** We collected detailed delivery dataset from Kaggle. And loaded the data set into a data frame using pandas.
2. **Data Exploration:** There are 45593 rows and 20 columns in the data frame. Those columns are

Numerical columns	Categorical columns
Time_taken(min)	Weatherconditions
Delivery_person_Age	City

Delivery_person_Ratings	Road_traffic_density
Restaurant_latitude	Type_of_vehicle
Restaurant_longitude	Festival
Delivery_location_latitude	
Delivery_location_longitude	
Order_Date	
Time_Orderd	
Time_Order_picked	
Vehicle_condition	
Type_of_order	
multiple_deliveries	

Column Name	Description
Time_taken(min)	Time taken by delivery person to complete the delivery
Delivery_person_Age	Age of the delivery person
Delivery_person_Ratings	Ratings of the delivery person
Restaurant_latitude	latitudinal location of the restaurant
Restaurant_longitude	longitudinal location of the restaurant
Delivery_location_latitude	latitudinal location of the delivery location
Delivery_location_longitude	longitudinal location of the delivery location
Order_Date	The date when the order is placed
Time_Orderd	The time when the order is placed
Time_Order_picked	The time when the order is picked
Vehicle_condition	The condition of the vehicle
Type_of_order	The type of order
multiple_deliveries	The number of deliveries
Weatherconditions	Condition of the weather
City	Type of the city
Road_traffic_density	Traffic condition
Type_of_vehicle	Type of the vehicle
Festival	Whether it is a festival day or not

- 3. Data Preparation:** Modifying the 'Time_taken', 'Weatherconditions', columns values by splitting them and extracting the first half of it. And creating additional column 'Code'.
- Dropping unnecessary columns that not required for this analysis like 'ID' and 'Delivery_person_ID'.
- Updated data types of the columns using a function named update_datatypes().
- Delivery_person_Age from object to float
- Delivery_person_Ratings from object to float
- multiple_deliveries from object to float
- Order_Date from object to datetime64[ns] format="%d-%m-%Y"
- Time_Orderd from object to timedelta64[ns]
- Time_Order_picked from object to timedelta64[ns]

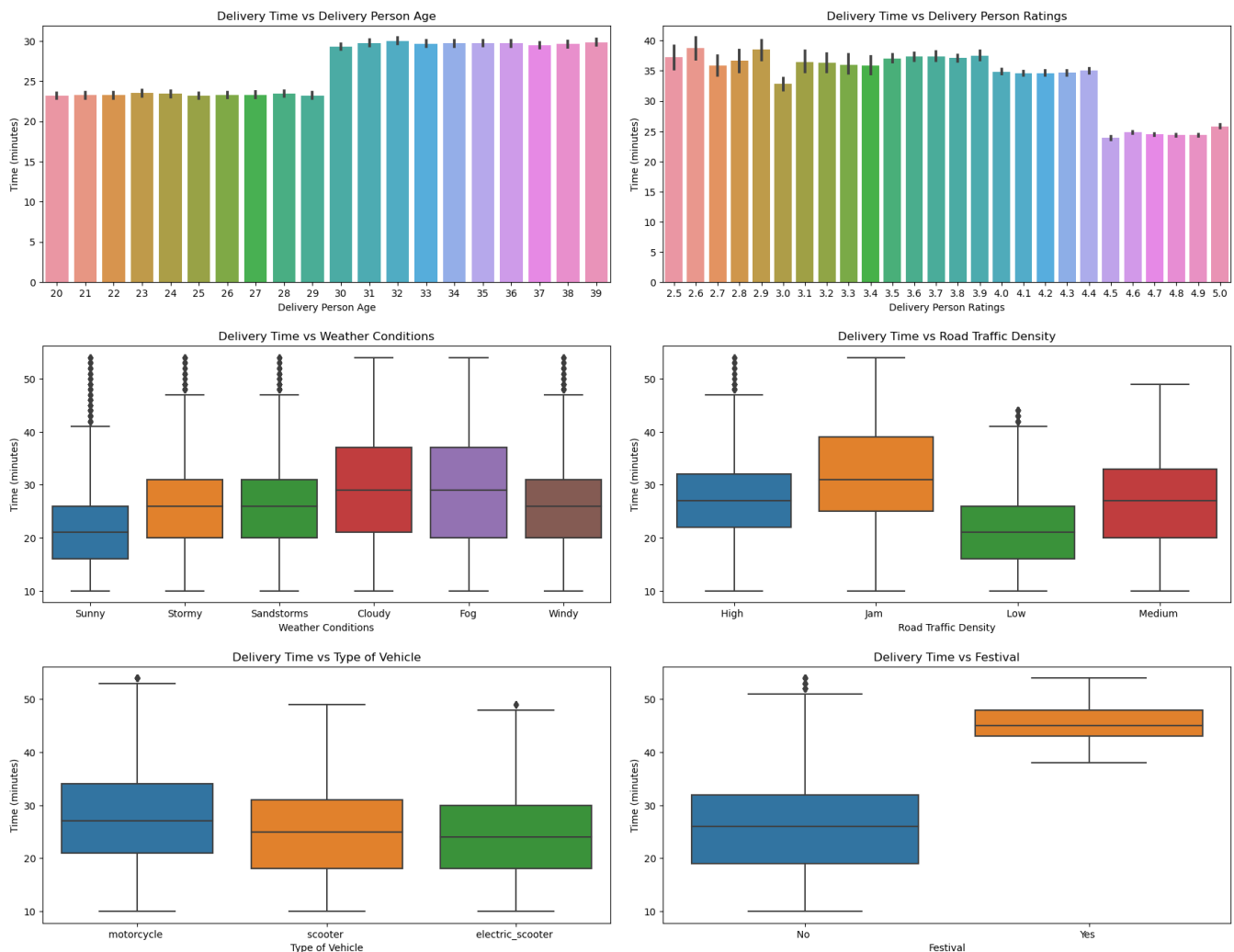
4. **Data Cleaning:** We cleaned the data by removing missing values and duplicates. There are 8 columns having null values. We have dropped the rows containing missing values using dropna() method.
5. **Data Visualization:** Using Python's Pandas and visualization libraries, We visualized the data into form of charts and graphs to identify key factors affecting delivery times. We have performed univariate, Bivariate and Multivariate Analysis.

Results:

Data Visualization:

Various visualization techniques illustrate the relationships and distributions within the data.

Can we build a predictive model to estimate delivery time based on a combination of weather conditions, road traffic density, and the type of vehicle used, while also considering the festival?



From the above subplots we can say that the age group below 30 years takes shorter time to deliver whereas the age group above 30 years takes more time comparatively.

The rating given is inversely proportional to the time taken for delivery. For a delivery person, to acquire a 4.5 rating the delivery must be completed within 25 minutes.

During the cloudy and foggy weather conditions the time taken to deliver is more comparatively to the other weather conditions.

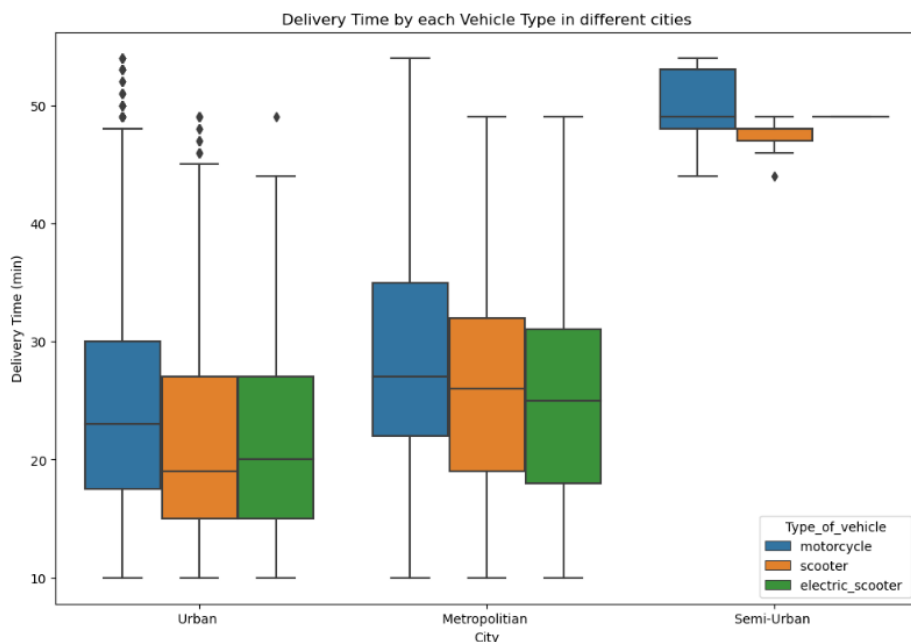
The traffic density is one of the major factors affecting delivery time as well.

The delivery person using electric scooter will delivery slightly more faster than the person using motorcycle and scooter.

During festival days the delivery time is rapidly increasing this may be due to the number of orders.

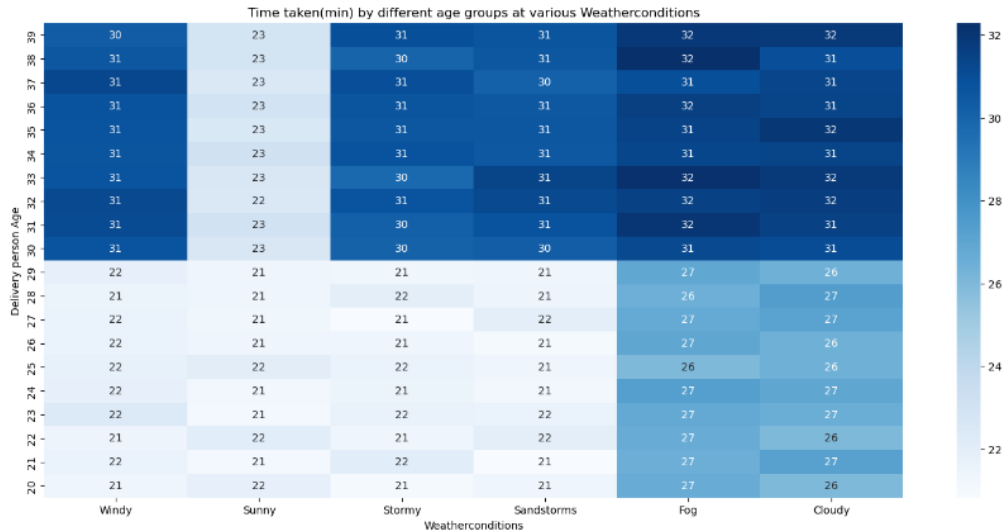
Delivery Time taken by each Vehicle Type in different cities

```
n [18]: # Boxplot of delivery time by city and type of vehicle
plt.figure(figsize=(12, 8))
sns.boxplot(x='City', y='Time_taken(min)', data=df, hue='Type_of_vehicle')
plt.title('Delivery Time by each Vehicle Type in different cities')
plt.xlabel('City')
plt.ylabel('Delivery Time (min)')
plt.show()
```



Motorcycles in all types of cities have more delivery time than scooters and electric scooters. Semi urban cities take more time than other cities types for the delivery. The box plot shows high deviation in urban and metropolitan type of cities which shows there is more data deviation in time taken in these cities.

```
In [27]: # Visualize as heatmap
plt.figure(figsize=(18, 8))
sns.heatmap(pivot1, annot=True, cmap="Blues")
plt.title('Time taken(min) by different age groups at various Weatherconditions')
plt.xlabel('Weatherconditions')
plt.ylabel('Delivery person Age')
plt.gca().invert_xaxis()
plt.gca().invert_yaxis()
plt.show()
```



The above heat maps shows that the delivery person age group below 30 are more likely to deliver faster than the age group above 30. Only in the sunny weather conditions both the age groups takes similar time whether as during other weather conditions age group below 30 are leading in fast deliveries.

Conclusion:

Our analysis effectively utilized data to forecast food delivery times. We identified critical factors such as vehicle type, weather, and traffic that impact the delivery speed. These insights will help improve delivery efficiency and customer satisfaction. The project's findings offer valuable strategies for food delivery companies to enhance their operations.

Future Scope:

In the future, we plan to improve this project by integrating real-time data on traffic and weather. We also aim to explore emerging technologies such as AI route optimization and drone deliveries. To make the model more accurate and applicable, we will expand the study to include more diverse geographic data. These advancements could significantly enhance our predictive capabilities and ensure that delivery services remain innovative and customer focused.

References:

- [1] Stack Overflow. (2019). Merge, join, concatenate, and compare.
<https://stackoverflow.com/questions/53645882/pandas-merging-101>
- [2] Pandas Tutorial:
<https://www.w3schools.com/python/pandas/default.asp>
- [3] Kaggle:
<https://www.kaggle.com/>